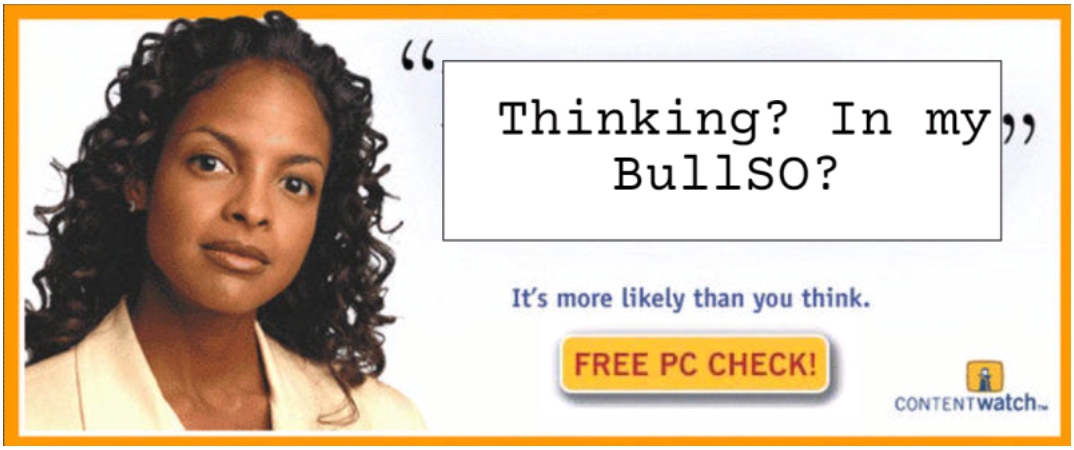
Dynamic Planet C - Dynamic Planet - Bulls Science Olympiad - 11-19-2022

Good luck!



1. (2.00 pts) Which of the following is not a meandering river?

- A) Uvac River
- B) Brahmaputra-Jamuna River
- C) Owens River
- D) Mississippi River

2. (2.00 pts) Which of the following rock types would not have pseudokarst sinkholes?

- A) Sandstone
- B) Limestone
- C) Gypsum
- D) Quartzite

3. (2.00 pts) =====

This symbol is used on topographic maps to represent what?

- A) River boundary
- B) National boundary
- C) Channel
- D) Road

4. (2.00 pts) ≈≈

This symbol is used on topographic maps to represent what?

- A) Perennial stream
- B) Perennial river
- C) Intermitten Stream
- D) Intermitten River

5. (2.00 pts) What size is silt?

- A) 1/256 - 1/16 mm
- B) <1/256 mm
- C) 1/16 - 2mm
- D) 2- 64 mm

6. (2.00 pts) Which of the following describes relative speeds of a meandering river correctly?

- A) Inside curve high and outside curve low
- B) Inside curve low and outside curve low
- C) Inside curve high and outside curve high
- D) Inside curve low and outside curve high

7. (2.00 pts) Which of the following described actions of the meandering river correctly?

- A) Inside curve erodes and outside curve deposits
- B) Inside curve deposits and outside curve deposits
- C) Inside curve erodes and outside curve erodes
- D) Inside curve deposits and outside curve erodes

8. (2.00 pts) What is wash load?

- A) The finest sediment in the suspended load
- B) The large sediment on the bottom of the river
- C) All the sediment deposited by the river
- D) All the sediment kept afloat by the river

9. (2.00 pts) What is stream capacity?

- A) The largest number of boats a stream can have
- B) The largest size sediment it can transport
- C) The total amount of sediment it can transport
- D) The total amount of water a stream can hold

10. (2.00 pts) Which of the following can describe slightly salty lakes

A) Saline
B) Brackish
C) Brine
D) Freshwater

11. (3.00 pts) What percent of Earth's water is stored in seas and oceans?

Expected Answer: Any number 96-97%

12. (4.00 pts) What is an ephemeral lake?

Expected Answer: a lake that is usually dry (2), but that fills with water for brief periods during and after rainfall or other precipitation. (2)

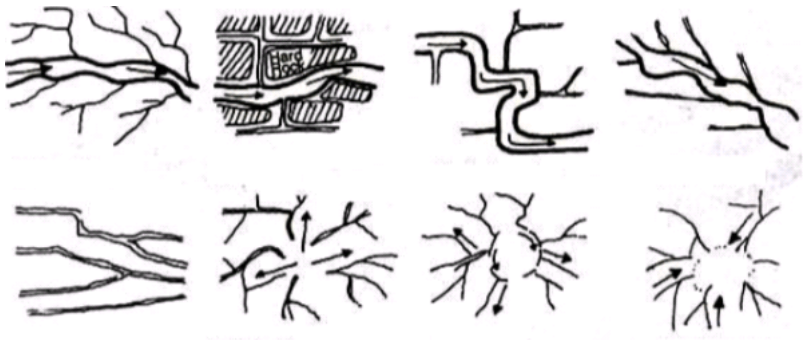
13. (7.00 pts) Identify the parts of a stratified lake (1 pt each) and explain what they are (1 pt each)

Expected Answer: Epilimnion - top warm layer Thermocline/Metalimnion (2 pts if both included (secret point)) - middle layer Hypolimnion - bottom cold layer

14. (2.00 pts) What is the the percentage of void space in a rock referred to as?

Porosity

15. (4.00 pts)



What are these called and how are they classified?

Expected Answer: Drainage Patterns (2pt) Classified on basis of form (shape) and texture

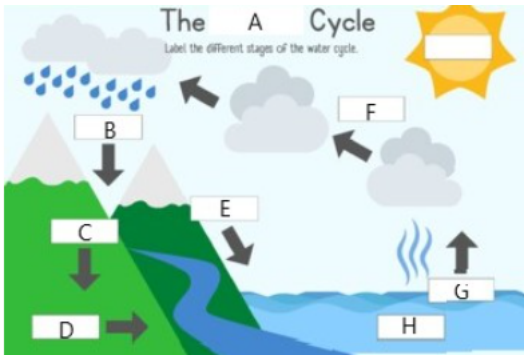
16. (8.00 pts) Referring to the previous image, classify them (top left to top right, then bottom left to bottom right)

Expected Answer: dendritic trellis rectangular angular parallel radial annular centripetal

17. (12.00 pts)
Bob goes on the Bull stream in Oklahoma. His speed, on average, is 7 mph, and it takes him 4 hours and 30 minutes to get to the end of the stream. His friend, Joe, starts at the same point on the stream but instead of using the stream walks straight west and straight north to the end of the stream. His speed is a constant 5 mph and he walks west for 3 hours and north for 4 hours. Based on this calculate the sinuosity of the stream and show work. Show work

Expected Answer: 7 mph * 4.5 hrs = 31.5 5 mph * 3 hrs = 15 miles 5 mph * 4 hrs = 20 miles 15^2 + 20^2 = 25^2 25 miles straight line distance 31.5/25 = 1.26 (no units = 2 pts)

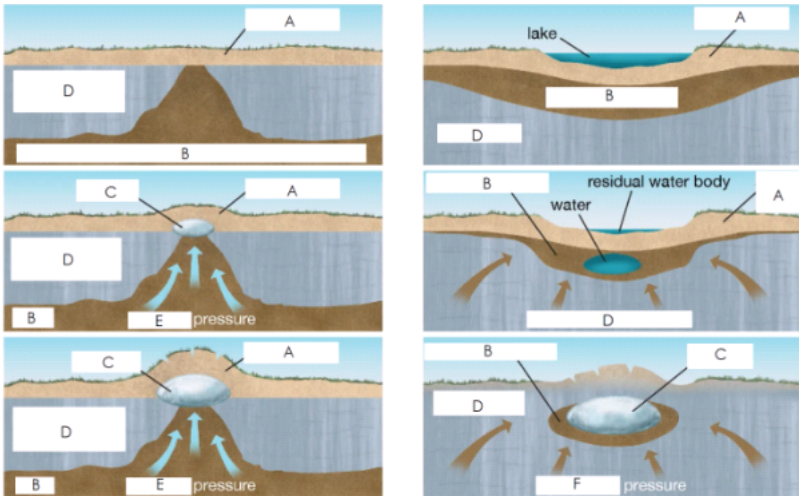
18. (8.00 pts)



Label the diagram: (C and D refer to the transfer through the mountain rather than on the surface)

Expected Answer: A: Hydrological/Water B: Precipitation C: Infiltration/percolation D: Groundwater flow E: Runoff F: Condensation G: Evaporation H: Ocean/Water body

19. (8.00 pts)



Label the diagram:

Expected Answer: A: Active layer B: Talik (unfrozen ground) C: Ice lens D: Permafrost E: Artesian F: Hydrostatic G: Open-system H: Closed-system

20. (2.00 pts) Which soil type contains the least hydraulic conductivity?

Expected Answer: Clay (accept any subcategory of clay)

For the below questions, answer with bog, marsh, or swamp:

21. (1.00 pts) Which is dominated by water-tolerant trees?

Swamp

22. (1.00 pts) Which is usually acidic?

Bog

23. (1.00 pts) Which has herbaceous vegetation?

Marsh

24. (1.00 pts) Which is generally located north and south of the tropics?

Marsh

25. (1.00 pts) Which is usually anoxic?

Bog

26. (1.00 pts) The water source of which one is mostly rivers and streams?

Swamps

27. (1.00 pts) The Florida Everglades are an example of what?

Marsh

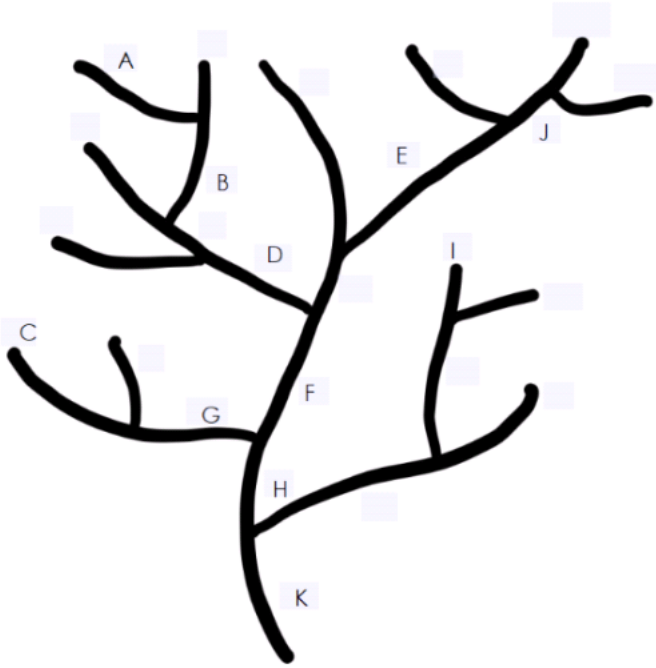
28. (2.00 pts) What is a grike?

Expected Answer: A fissure made of limestone enlarged by dissolution from rainwater

29. (2.00 pts) What is the name of a cutoff meander loop?

Expected Answer: Oxbow lake

30. (11.00 pts)



Label the stream order of each stream in the diagram above with the Strahler method:

A:

B:

C:

D:

E:

F:

G:

H:

I:

J:

K:

Expected Answer: A: 1 B: 2 C: 1 D: 2 E: 2 F: 3 G: 2 H: 3 I: 1 J: 2 K: 3

31. (11.00 pts) Now use the Shreve method:

A:

B:

C:

D:

E:

F:

G:

H:

I:

J:

K:

Expected Answer: A: 1 B: 2 C: 1 D: 4 E: 3 F: 8 G: 2 H: 10 I: 1 J: 2 K: 13

32. (2.00 pts) What is an esker?

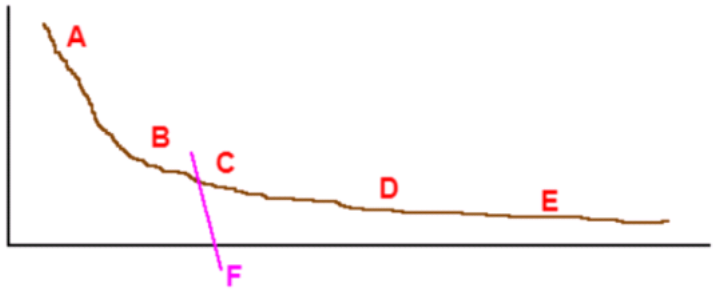
Expected Answer: A long ridge of graver and sediment deposited by meltwater from a retreating glacier or ice sheet

33. (12.00 pts) Model a channel so that it is represented by a semicircle with radius 1 m. Assume it is filled completely through the brim. Show all work for the following.

1. What is the wetted perimeter? (2)
2. What is the hydraulic radius? (2)
3. The channel travels 1000 m and lowers 2 m in elevation. What is its gradient? (2)
4. Using a Chezy coefficient of 20 m^(½)/s, calculate the velocity (2)
5. What is the volumetric flow rate? (2)
6. From the Chezy coefficient, characterize the type of channel it is in terms of size and roughness. (2)

Expected Answer: 1. Pi x r = 3.14 m (1 for answer, 1 for work), 2. 0.5 x pi x r^2 / pi / r = 0.5r = 0.5 m(1 for answer, 1 for work), 3. 2/1000 = 0.002 (1 for answer, 1 for work), 4. 20 * sqrt(0.5 x 0.002) = .632 (1 for answer, 1 for work), 5. 0.5 x pi x r^2 x prev answer = .993 (1 for answer, 1 for work), 6. Small (1) and rough (1)

34. (8.00 pts)



1. This image shows a profile of a stream at fluvial equilibrium. Give the letters (A-E) for points that match the following: (4)

Highest elevation

Have more bedrock character

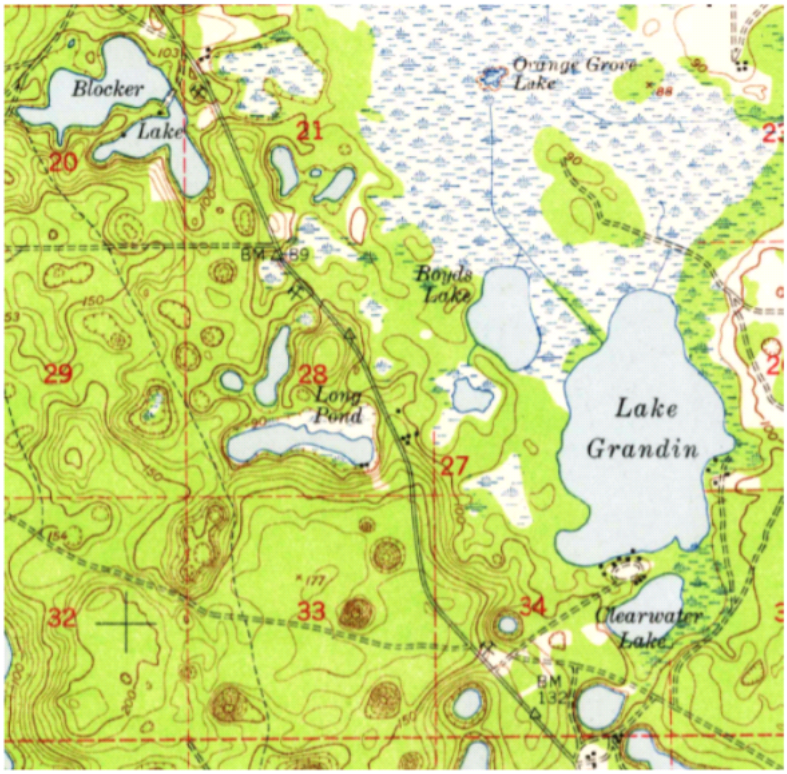
Have the lowest velocity

Contain generally large clasts.

2. Referring to the same image, describe how a normal fault occurring at F would effect rates of deposition/erosion at points B/C. (4)

Expected Answer: AAEA (1 each), At point C (1) deposition increases (1) At point B (1) erosion increases (1)

35. (8.00 pts)



1. Identify the primary Karst topographic feature shown around the 33 and 34 markers (2)

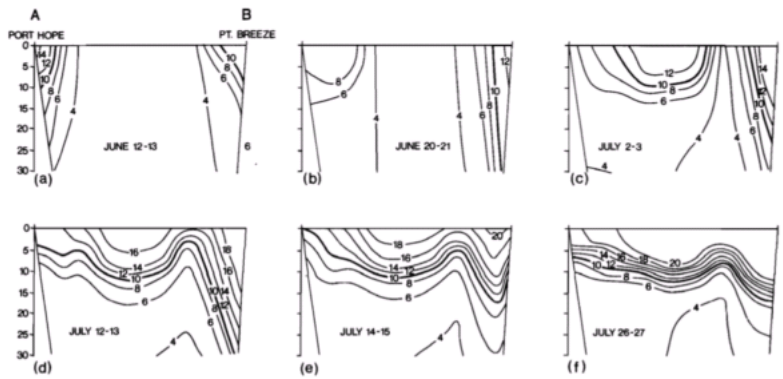
2. Propose a mechanism as to how the small circular ponds west of Clearwater lake formed. (2)

3. What is the contour interval of the map? Map units are in feet (2)

4. What is the Cvijic classification of the Karst process this map is in? Explain how you know (2)

Expected Answer: 1. Sinkhole (2), 2. Sinkholes (1) hit the water table (1), 3. 10 ft (2), 4. Mature (1) many holes (1)

36. (4.00 pts)



Thermal structure of Lake Ontario

1. Describe what process is occurring in the thermal stratification from (a) to (c)

2. Describe what heating process is most responsible for the changes in the thermal regime from (c) to (f)

3. Where do you expect to find the highest density water in map (f)? How do you know? (2)

Expected Answer: 1. Upwelling (1), 2. Differential solar heating (1), 3. Lower center (1) due to low temperature (1)

37. (5.00 pts)



This is part of Indiana's dry season.

1. What is the lowest gage height achieved by the creek in ft?

2. What is the classification of this stream (perennial, ephemeral, or intermittent)? Explain how you know

3. Would this data be able to inform the highest annual river stage? Explain why or why not?

Expected Answer: 1. 1 ft (1), 2. Perennial (1) description (1), 3. No (1) because dry (1)

38. (8.00 pts) 1. Assuming cubic packing of spherical particles in a rock, where the unit cube has a center of a particle at each vertice, calculate the porosity as a %. Show work!

2. Would this rock be a good aquifer or a good aquiclude? How do you know?

3. You fill a tube with this rock and run water through it. You notice more Darcian flux when you tilt the tube more. What accounts for this effect, and why does it occu

Expected Answer: 1. (s^3 - 4/3 * π * (s/2)^3) / s^3 = 0.4764 = 47.64% (2 for work, 1 for answer), 2. Good aquifer (1) very porous (1), 3. Pressure (1) some explanation that makes sense (2)

39. (16.00 pts) Name as many types of lakes as you can.

Expected Answer: Meteorite (Crater) Anthropogenic Shoreline Aeoline Solution Landslide Fluvial Tectonic Glacial Volcanic Organic Salty Artifical Oxbow Meromictic Rift Ephemeral Stratified

40. (6.00 pts) Explain characteristics of Young, Mature, and Old Rivers

Expected Answer: Young- fast flow, erode faster, located in high elevations Mature- change erosion patterns, less vertical erosion/more horizontal, meanders as closer to ground Old- final stage- many meanders, slowest stage, flows almost ground level, deposition occurs- at bottom of river, creates oxbow lakes

Congratulations on finishing the Dynamic Planet C Test!

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